

NPTEL
NPTEL ONLINE CERTIFICATION COURSE
Reinforcement Learning
Week: 0
Lecture: 1
Tutorial 1 Probability Basic-1
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Hello and welcome to the first tutorial in the introduction to machine learning course. My name is Priyatosh. I am one of the teaching assistants for this course. In this tutorial we will be looking at some of the basics of probability theory. Before we start let us discuss the objectives of this tutorial. The aim here is not to teach the concepts of probability theory in any great detail instead we will just be providing a high level overview of the concepts that we will encounter later on in the course the idea here is that for those of you have a done course on probability theory or are otherwise familiar with the content this tutorial should act as a refresher.

For others who may find some of the concepts unfamiliar we recommend that you go back and prepare those concepts from say an introductory text book or any other resource so that when you encounter those concepts later on in the course you should be comfortable with that okay.

(Refer Slide Time: 01:12)

Sample Space

Sample Space: The set of all possible outcomes of an experiment is called the sample space and is denoted by Ω . Individual elements are denoted by ω and are termed elementary outcomes.

Examples:

- ▶ (Finite) A single roll of an ordinary die. Here, $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- ▶ (Countable) Infinite number of coin tosses in order to study, say, the number of tosses before 5 consecutive heads are observed. Here, $\Omega = \{H, T\}^\infty$.
- ▶ (Uncountable) Speed of a vehicle measured with infinite precision. Here, $\Omega = \mathbb{R}$.



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To start this tutorial we will look at some of the technicians of some of the fundamental concepts. The first one to consider is that of the sample space the set of all possible outcomes of an experiment is called the sample space and it is denoted by Ω . Individual elements are denoted by ω and are termed elementary outcomes let us consider some examples in the first example the experiment consists of rolling an ordinary die the sample space here is a set of numbers between one and six each individual element here represents one of the six possible outcomes of rolling a die.

Note that in this example the sample space is finite in the second example the experiment consists of tossing a coin repeatedly until a specified condition is observed, here we are looking to observe five consecutive heads before terminating the experiment, the sample space here is countably infinite we the individual elements are represented using a sequence of H and T is where H and T stand for Heads and Tails respectively in the final example the experiment consists of measuring the speed of the vehicle with infinite precision.

Assuming that the vehicle speeds can be negative the sample space is clearly the set of real numbers here we observe that the sample space can be uncountable.

(Refer Slide Time: 02:34)

Set Theory Notations

$$A \subset B \Leftrightarrow x \in A \Rightarrow x \in B$$

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$A^c = \{x : x \notin A\}$$



The next concept we look at is that is of an event. An event is any collection of possible outcomes of an experiment that is any subset of the sample space. The reason why events are important to us is because in general when we conduct an experiment we are not really that interested in the elementary outcomes rather we are more interested in some sub sets of the elementary outcomes. For example on rolling a die we might be interested in observing whether the outcome was even or odd.

So for example on a specific roll of a die let us say we observed that the outcome was odd. In this scenario whether the outcome was actually a one or three or a five is not as important to us as the fact that it was odd since we are considering sets in terms of sample spaces and events we will quickly go through the basic set theory notations. As usual capital letters indicates sets and small letters indicate set elements.

We first look at the subset relation for all x if $x \in A \implies x \in B$ then we say that A is a subset of B or A is contained in B . Two sets A and B are said to be equal if both $A \subset B$ and $B \subset A$ holds. The union of two sets A and B gives rise to a new set which contains elements of both A and B , similarly the intersection of two sets gets rise to a new set which contains of only those elements which are common to both A and B .

Finally the complement of the set A is essentially the set which contains all elements in the universal set except for the elements present in A . In our case the universal set is essentially the sample space.

(Refer Slide Time: 04:25)

Properties of Set Operations

Commutativity

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associativity

$$A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

Distributivity

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$



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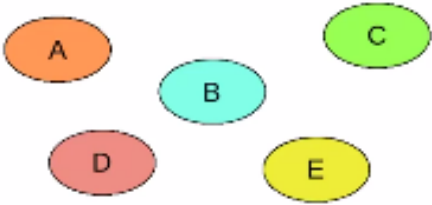
This slide lists out the different properties of set operation such as commutativity, associativity and distributivity which you should all be familiar with. It also list out the DeMorgan's Laws which can be very useful. According to DeMorgan's Laws $(A \cup B)^c = A^c \cap B^c$. Similarly $(A \cap B)^c = A^c \cup B^c$. The DeMorgan's laws presented here are for two sets, they can easily be extended for more than two sets.

(Refer Slide Time: 04:59)

Disjoint Events

Two events A and B are disjoint (or mutually exclusive) if $A \cap B = \phi$.

A sequence of events $A_1, A_2, A_3, \dots$ are pair-wise disjoint if $A_i \cap A_j = \phi$ for all $i \neq j$.



The diagram illustrates five disjoint events represented by colored ovals: A (orange), B (cyan), C (green), D (red), and E (yellow). The ovals are arranged such that no two ovals overlap, visually demonstrating that the intersection of any two events is an empty set.

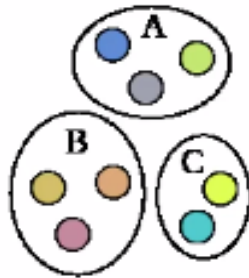
 

Coming back to events, two events A and B are said to be disjoint or mutually exclusive if the intersection of two sets is empty. Extending this concept to multiple sets we say that a sequence of events A_1, A_2, A_3 and so on are pair wise disjoint if $A_i \cap A_j = \phi$ for all $i \neq j$. In the example below if each of the letters represents an event then the sequence of the events A through E are pairwise disjoint since the intersection of any pair is empty.

(Refer Slide Time: 05:35)

Partition

If $A_1, A_2, \dots$ are pair-wise disjoint and $\bigcup_{i=1}^{\infty} A_i = \Omega$, then the collection $A_1, A_2, \dots$ forms a partition of Ω .



If events A_1, A_2, A_3 so on are pair wise disjoint and the union of the sequence of events gives rise to the sample space then the collection A_1, A_2 and so on is said to form a partition of the sample space Ω , this is illustrated in the figure below.

(Refer Slide Time: 05:57)

Sigma Algebra

Given a sample space Ω , a σ -algebra is a collection $\mathcal{F}$ of subsets of Ω , with the following properties:

- (a) $\Phi \in \mathcal{F}$.
- (b) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.
- (c) If $A_i \in \mathcal{F}$ for every $i \in \mathbb{N}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

A set A that belongs to $\mathcal{F}$ is called an $\mathcal{F}$ -measurable set (event).

Example: Consider $\Omega = \{1, 2, 3\}$.

$\mathcal{F}_1 = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.

$\mathcal{F}_2 = \{\emptyset, \{1, 2, 3\}\}$.



Next we come to the concept of a σ -algebra given a sample space Ω , a σ -algebra is a collection $\mathcal{F}$ of subsets of the sample space with the following properties. The null set is an element of $\mathcal{F}$, if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$. If $A_i \in \mathcal{F} \forall i \in \mathbb{N}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$. A set A that belongs to $\mathcal{F}$ is called as a $\mathcal{F}$ -measurable set. This is what we naturally understand as an event.

So going back to the third property what this essentially says is that if there are a number of events which belong in the set σ -algebra then the countable union of these events also belongs in the σ -algebra, let us consider an example consider a $\Omega = \{1, 2, 3\}$. This is our sample space. With this sample space we can construct a number of different σ -algebras. Here the first σ -algebra $\mathcal{F}_1$ is essentially the power set of the sample space.

All possible events are present in the first σ -algebra. However if we look at $\mathcal{F}_2$ in this case there are only two events the null set or the sample space itself, you should verify that for both $\mathcal{F}_1$ and $\mathcal{F}_2$ all

three properties listed above are satisfied. Now that we know what a σ -algebra is let us try and understand how this concept is useful.

(Refer Slide Time: 07:34)

Sample Space Size Considerations

For any Ω (countable or uncountable) 2^Ω is always a σ -algebra.

For example, for $\Omega = \{H, T\}$, a feasible σ -algebra is the power set, i.e., $\mathcal{F} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$.

However, if Ω is uncountable, then probabilities cannot be assigned to every subset of 2^Ω .



First of all for any Ω , countable or uncountable the power set is always a σ -algebra. For example for the sample space comprising of two elements H, T a feasible σ -algebra is the power set. This is not the only feasible σ -algebra as we have seen in the previous example. But always the power set will give you a feasible σ -algebra, however if Ω is uncountable, the probabilities cannot be assigned to every subset of the power set.

This is the crucial point which is why we need the concept of σ -algebras. So just to recap if the sample space is finite or countable then we can kind of ignore the concept of σ -algebra because in such a scenario we can consider all possible events, that is the power set of the sample space and meaningfully apply probabilities to each of these events. However this cannot be done when the sample space is uncountable that is if Ω is uncountable, then probabilities cannot be assigned to every subset of 2^Ω .

This is where the concept of σ -algebra shows its use when we have an experiment in which the sample space is uncountable. For example, let us say the sample space is the set of real numbers. In

such a scenario we have to identify the events which are of importance to us and use this along with the 3 properties listed in the previous slide to construct a σ -algebra and probabilities will then be assigned to the collection of sets in the σ -algebra.

(Refer Slide Time: 09:31)

Probability Measure & Probability Space

A probability measure $\mathcal{P}$ on $(\Omega, \mathcal{F})$ is a function $\mathcal{P} : \mathcal{F} \rightarrow [0, 1]$ satisfying

(a) $\mathcal{P}(\phi) = 0$, $\mathcal{P}(\Omega) = 1$;

(b) if $A_1, A_2, \dots$ is a collection of pair-wise disjoint members of $\mathcal{F}$, then

$$\mathcal{P}(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathcal{P}(A_i)$$

The triple $(\Omega, \mathcal{F}, \mathcal{P})$, comprising a set Ω , a σ -algebra $\mathcal{F}$ of subsets of Ω , and a probability measure $\mathcal{P}$ on $(\Omega, \mathcal{F})$, is called a **probability space**.



Next we look at the important concepts of probability measure and probability space. A probability measure P on a specific sample space Ω and σ -algebra $\mathcal{F}$ is a function from $\mathcal{F}$ to $[0, 1]$ satisfies the following properties. $P(\phi) = 0$; $P(\Omega) = 1$ and if $A_1, A_2, \dots$ is a collection of pair wise disjoint members of $\mathcal{F}$ and probability of the union of all such members is equals to the sum of their individual probabilities.

Note that this holds because the sequence $A_1, A_2, \dots$ is pair wise disjoint. The triple $(\Omega, \mathcal{F}, P)$ comprising a sample space Ω , a σ -algebra $\mathcal{F}$ which are subsets of Ω and a probability measure P defined on $\Omega, \mathcal{F}$. This is called a probability space, for every probability problem that we come across there exists a probability space comprising of the triple $(\Omega, \mathcal{F}, P)$, even though we may not always explicitly take

into consideration this probability space when we solve the problem it should always remain in the back of our heads.

(Refer Slide Time: 10:45)

Example

Consider a simple experiment of rolling an ordinary die in which we want to identify whether the outcome results in a prime number or not.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$
$$\mathcal{F} = \{\phi, \{1, 4, 6\}, \{2, 3, 5\}, \{1, 2, 3, 4, 5, 6\}\}$$
$$\mathcal{P} : \mathcal{F} \rightarrow [0, 1]$$

- ▶ $\mathcal{P}(\phi) = 0$
- ▶ $\mathcal{P}(\{1, 4, 6\}) = 0.5$
- ▶ $\mathcal{P}(\{2, 3, 5\}) = 0.5$
- ▶ $\mathcal{P}(\Omega) = 1$



Let us now look at an example where we do consider the probability space involved in the problem. Consider a simple experiment of rolling an ordinary die in which we want to identify whether the outcome results in a prime number or not, the first thing to consider is the sample space. If there are only six possible outcomes in our experiment, the sample space here is consists of the numbers between 1 to 6. Next we look at the σ –algebra. Note that since the sample space is finite you might as well consider all possible events that is the power set of the sample space.

However note that the problem dictates that we are only interested in two possible events that is whether the number, that is whether the outcome is prime or not. Thus restricting ourselves to these two events we can construct a simpler σ –algebra. Here we have two events which correspond to the events we are interested in and the remaining two events follow from the properties which a σ –algebra has to follow. Please verify that the σ –algebra listed here does actually satisfy the 3 properties that we have discussed above. The final component is the probability measure. The probability measure assigns value between 0 and 1 that is the probability value to each of the components of the σ –algebra.

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Example

Consider a simple experiment of rolling an ordinary die in which we want to identify whether the outcome results in a prime number or not.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$
$$\mathcal{F} = \{\emptyset, \{1, 4, 6\}, \{2, 3, 5\}, \{1, 2, 3, 4, 5, 6\}\}$$
$$\mathcal{P} : \mathcal{F} \rightarrow [0, 1]$$

- ▶ $\mathcal{P}(\emptyset) = 0$
- ▶ $\mathcal{P}(\{1, 4, 6\}) = 0.5$
- ▶ $\mathcal{P}(\{2, 3, 5\}) = 0.5$
- ▶ $\mathcal{P}(\Omega) = 1$



Here the values listed assumes that, the die is fair in the sense that, the probability of each face is equals to $1/6$. Having covered some of the very basics of probability, in the next few slides we look at some rules which allow us to estimate probability values.

(Refer Slide Time: 12:22)

Bonferroni's Inequality

$$P(A \cap B) \geq P(A) + P(B) - 1$$

General form:

$$P(\cap_{i=1}^n A_i) \geq \sum_{i=1}^n P(A_i) - (n-1)$$



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The first thing we look at is known as the Bonferroni's Inequality. According to this inequality, the $P(A \cap B) \geq P(A) + P(B) - 1$, the general form of this inequality is also listed. What this inequality allows us to do is to give a lower bound on the intersection probability; this is useful when the intersection probability is hard to calculate.

(Refer Slide Time: 12:56)

Bonferroni's Inequality

$$P(A \cap B) \geq P(A) + P(B) - 1$$

General form:

$$P(\cap_{i=1}^n A_i) \geq \sum_{i=1}^n P(A_i) - (n-1)$$

Gives a lower bound on the intersection probability which is useful when this probability is hard to calculate.

Only useful if the probabilities of individual events are sufficiently large.



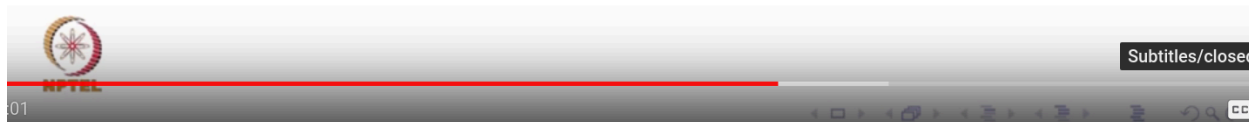
However if you notice the right hand side of the inequality, you should observe that this result is only meaningful when the individual probabilities are sufficiently large. For example, if the $P(A)$ and the $P(B)$, both these values are very small, then this -1 term dominates and the result doesn't make much sense.

(Refer Slide Time: 13:16)

Boole's Inequality

$$P(\cup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i), \text{ for any sets } A_1, A_2, \dots$$

Gives a useful upper bound for the probability of the union of events.



According to the Boole's inequality, for any sets, A_1, A_2 and so on, the probability of the union of these sets is always less than equals to the sum of their individual probabilities. Clearly this gives us a useful upper bound for the probability of the union of events. Notice that this equality only will hold, when these sets are pairwise disjoint.

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Conditional Probability

Given two events A and B , if $P(B) > 0$, then the conditional probability that A occurs given that B occurs is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



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Next we look at conditional probability, given two events A and B , if $P(B) > 0$, then the conditional probability that A occurs, given that B occurs is defined to be the $P(A|B) = \frac{P(A \cap B)}{P(B)}$

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Conditional Probability

Given two events A and B , if $P(B) > 0$, then the conditional probability that A occurs given that B occurs is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Essentially, since event B has occurred, it becomes the new sample space.

Conditional probabilities are useful when reasoning in the sense that once we have observed some event, our beliefs or predictions of related events can be updated/improved.



Essentially, since event B has occurred, it becomes the new sample space and now the probability of A is accordingly modified. Conditional probabilities are very useful when reasoning. In the sense that once we have observed some event our beliefs or predictions of related events can be updated or improved. Let us try working out a problem in which conditional probabilities are used, A fair coin is tossed twice, what is the probability that both tosses result in heads given that at least one of the tosses resulted in a heads?

Go ahead and pause the video here and try working out the problem yourself. From the question it is clear that, that there are four elementary outcomes, both tosses resulted in heads, both came up tails, the first came up heads while the second toss came up tails and the other way round. Since we are assuming that the coin is fair, each of the elementary outcomes have the same probability of occurrence equals to $1/4$. Now we are interested in the probability that both tosses come up heads, given that at least one resulted in a head.

Applying the conditional probability formula, we have $P(A|B)$ equals to $P(A \cap B)/P(B)$. Simplifying the intersection in the numerator, we get the next step. Now we can apply the probability values of the elementary outcomes to get the result of $1/3$, note that in the denominator, each of these events is mutually exclusive, thus the probability of union of these three events, is equals to the summation of the individual probabilities.

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Example

Q. A fair coin is tossed twice. What is the probability that both tosses result in heads given that at least one of the tosses resulted in a heads?

Sol. $\Omega = \{HH, TT, HT, TH\}$
 $P(HH) = P(TT) = P(HT) = P(TH) = 1/4$

$P(HH | \text{at least one toss heads})$
 $= \frac{P(HH \cap (HT \cup TH \cup HH))}{P(HT \cup TH \cup HH)}$
 $= \frac{P(HH)}{P(HT \cup TH \cup HH)}$
 $= \frac{1/4}{3/4}$
 $= \frac{1}{3}$



As an exercise try and solve the same problem with the modification that we observe only the first toss coming up heads that is we want the probability that both tosses resulted in heads given that the first toss resulted in a heads, does this change the problem? Next we come to a very important theorem called the Bayes's theorem or the Bayes's Rule; we start with the equation for the conditional probability.

$$P(B) = \frac{P(A \cap B)}{P(A)}$$

Rearranging we have $P(A \cap B) = P(A)P(B)$, now instead of starting with $P(A|B)$, if I started with $P(B|A)$, you would have got $P(A \cap B) = P(B)P(A)$. These two right hand sides can be equated to get $P(A)P(B) = P(B)P(A)$. Now taking this $P(B)$ to the right hand side we get, $P(A) = \frac{P(A \cap B)}{P(B)}$. This is what is known as the Bayes' rule.

(Refer Slide Time: 17:20)

Bayes' Rule

We have:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A|B)P(B)$$

$$P(A \cap B) = P(B|A)P(A)$$

$$P(A|B)P(B) = P(B|A)P(A)$$

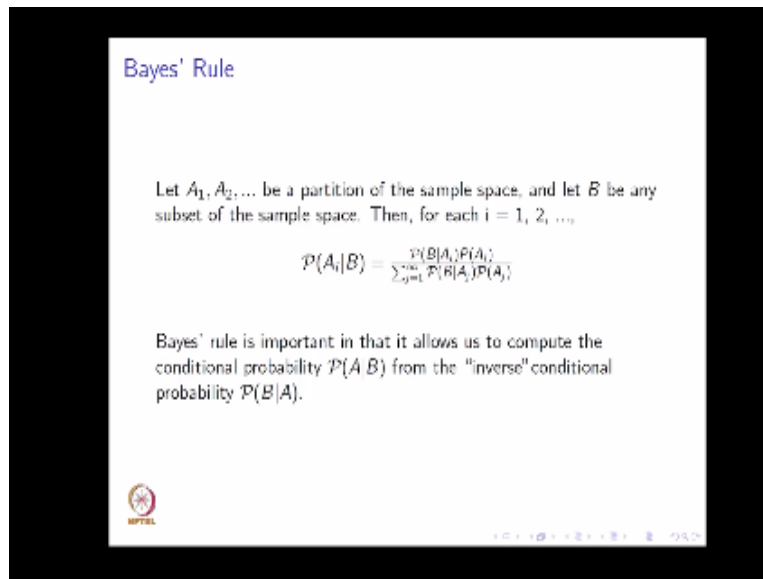
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (\text{Bayes' Rule})$$



9

Note that what it essentially says is, if I want to find the probability of A, given that B happened, I can use the information of $P(B|A)$ along with the knowledge of $P(A)$ and $P(B)$ to get this value. As we will see this is a very important formula, here we again present the Bayes' rule in its expanded form where A_1, A_2 and so on form a partition of the sample space. As mentioned Bayes' rule is important in that it allows us to compute the conditional probability of A given B, from the inverse conditional probability of B given A.

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Bayes' Rule

Let $A_1, A_2, \dots$ be a partition of the sample space, and let B be any subset of the sample space. Then, for each $i = 1, 2, \dots$,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{\infty} P(B|A_j)P(A_j)}$$

Bayes' rule is important in that it allows us to compute the conditional probability $P(A|B)$ from the "inverse" conditional probability $P(B|A)$.

Let us look at a problem where Bayes' rule is applicable. To answer a multiple choice question a student may either know the answer or may guess it, assume that with probability P the student knows the answer to the question and with probability Q the student guesses the right answer to the question she does not know, what is the probability that, for a question the student answers correctly, she actually knew the answer to the question? Again pause the video here and try solving the problem yourself.

Let us first assume that K is the event that the student knows the question and C be the event that the student answers the question correctly. Now from the question we can gather the following information, the probability that the student knows the question is p , hence the probability that the student does not know the question is $1-p$, The probability that the student answers the question correctly, given that she knows the question is equals to 1, because if she knows the question, she will definitely answer it correctly. Finally the probability that the student answers the question correctly, in which she makes the guess, that is she does not know the question is q .

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Example

Q. To answer a multiple choice question, a student may either know the answer or may guess it. Assume that with probability p the student knows the answer to a question, and with probability q , the student guesses the right answer to a question she does not know. What is the probability that for a question the student answers correctly, she actually knew the answer to the question?

Sol. Let K be the event that the student knows the question, and C be the event that the student answers the question correctly.

We have $P(K) = p$, $P(-K) = 1 - p$, $P(C|K) = 1$, $P(C|-K) = q$

$$\begin{aligned} P(K|C) &= \frac{P(C|K)P(K)}{P(C)} \\ &= \frac{1 \cdot p}{p + q(1-p)} \end{aligned}$$



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We are interested in the probability of the student knowing the question given that she has answered it correctly. Applying bayes' rule we have $P(K|C) = \frac{P(C|K)P(K)}{P(C)}$. The probability of answering the question correctly can be expanded in the denominator to consider the two situations, probability of answering the questions correctly given that, the student knows the question and probability of answering the question correctly, given that the student does not know the question. Now using the values which we have gathered from the question, we can arrive at the answer of $\frac{p}{p+q(1-p)}$

Note here that the bayes' rule is essential to solve this problem, because while from the question itself, we have a handle on this value $P(C|K)$, there is no direct way to arrive at the value of $P(K|C)$. Two events A and B are said to be independent if $P(A \cap B) = P(A)P(B)$. More generally a family of events A_i , where i is an element of integers, is called independent, if probability of some subset of the events A_i is equals to the product of the probabilities of those events. Essentially what we are trying to say here is that, if you have a family of events A_i , then the independence condition holds only if for any subset of those events, this condition holds.

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Independent Events

Two events, A and B , are said to be independent if

$$P(A \cap B) = P(A)P(B)$$

More generally, a family $A_i : i \in I$ is called independent if

$$P(\cap_{i \in J} A_i) = \prod_{i \in J} P(A_i)$$

for all finite subsets J of I .

From the above, it should be clear that pair-wise independence does not imply independence.

*





From this it should be clear that pairwise independence does not imply independence. That is, pairwise independence is a weaker condition.

Extending the notion of independence of events, we can also consider conditional independence. Let A , B and C be three events with $P(C) > 0$, the event A and B are called conditionally independent given C if $P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)}$. This condition is very similar in form to the previous condition for independence of events, equivalently the events A and B are conditionally independent given C if, $P(A \cap B | C) = P(A | C)P(B | C)$. This latter condition is quite informative, what it says is that the probability of A calculated after knowing the occurrence of event C is same as the probability of A , calculated after having knowledge of occurrence of both events B and C .

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Conditional Independence

Let A , B , and C be three events with $P(C) > 0$. The events A and B are called conditionally independent given C if

$$P(A \cap B | C) = P(A | C)P(B | C)$$

or equivalently

$$P(A | B \cap C) = P(A | C)$$

Example: Assume that admission into the M.Tech. programme at IITM & IITB is based solely on candidate's GATE score. Then

$$P(IITM | IITB \cap GATE) = P(IITM | GATE)$$



Thus observing the occurrence or non-occurrence of B does not provide any extra information and thus we can conclude that the events A and B are conditionally independent, given C . Let us consider an example, assume that admission into the M tech programmes at IIT Madras and IIT Bombay, is based solely on the candidate's GATE score, then probability of admission into IIT Madras, given knowledge of the candidates admission status in IIT Bombay

As well as the candidate's GATE score is equivalent to the probability calculated simply knowing the candidate's GATE score. Thus knowing the status of the candidate's admission to IIT Bombay does not provide any extra information, hence, since the condition is satisfied, we can say that admission into the programme at IIT Madras, and admission into the programme at IIT Bombay are independent events given knowledge of the candidate's GATE score

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